

Deepu James

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Experience

Centre for Development of Telematics (C-DOT)

Jun – Jul 2025

New Delhi

Software Engineering Intern

- Automated diagnostic log analysis via Bash scripting, cutting manual review time by **70%**.
- Improved embedded error-flag detection pipeline, raising fault-detection accuracy by **35%**.

VDocs

Jun – Jul 2025

Remote

Web Developer Intern

- Re-architected frontend asset loading pipeline, improving page-load speed by **28%**.
- Integrated REST APIs; raised WCAG accessibility score from **62 to 88**.

Projects

AgriAdvisor — Agricultural LLM Chatbot

QLoRA • FAISS • RAG • Gradio • Hugging

Face

- Built end-to-end LLM chatbot with QLoRA PEFT fine-tuning and FAISS-based retrieval-augmented generation.
- Evaluated hallucination reduction, retrieval quality, and PEFT efficiency against baselines.

CBAM-YOLOv8n — Cassava Disease Detection

PyTorch • YOLOv8 • CBAM Attention •

TFLite

- Integrated CBAM attention into YOLOv8n; improved **mAP50 from 0.4623 to 0.4823**.
- Compressed model from **45 MB to 12 MB** while sustaining **15–30 FPS** on-device.

VoiceMed Assist — AI Clinical Scribe

Whisper • ECAPA-TDNN • Flask • Diarization

- Built bilingual speech-to-text pipeline for doctor–patient consultations using Whisper.
- Extracted symptoms, diagnoses, and treatments into structured Digital Health Records.

Real-Time Bully Detection Chat App

Python • TensorFlow Lite • NLP • Quantization

- Deployed quantized on-device NLP classifier with **92% toxicity accuracy** and **28 ms latency**.

Digital Twin Simulation — EV Battery Network

React • Node.js • WebSockets • Python

- Led platform UI development for real-time EV battery State-of-Health visualization.
- Designed WebSocket pipeline targeting **20% downtime reduction**.

QLoRA Fine-Tuning — Gemma-2-2b-it

Hugging Face • 4-bit QLoRA • ROUGE/BLEU

- Fine-tuned Gemma-2-2b-it locally on RTX 3060 (6 GB VRAM) using 4-bit QLoRA.

FireSmokeNet

PyTorch • CNN • D-Fire Dataset

- Designed custom CNN for fire/smoke/neutral classification with IEEE-style documentation.

Education

BML Munjal University

B.Tech CSE (Data Science & AI)

2023 – 2027 • Gurgaon

CGPA: 7.3 / 10

Coursework: ML, Deep Learning, DSA, System Design, Databases

Skills

Languages: Python, JavaScript, SQL, C++, Java, Shell

AI / ML: PyTorch, TensorFlow, Hugging Face, QLoRA, FAISS, RAG, Whisper, YOLOv8, ECAPA-TDNN, OpenCV, Scikit-learn

Web / Infra: React, Flask, FastAPI, Node.js, Express, Docker, Git, Linux, MySQL, MongoDB

Hardware: Arduino, ESP32, Raspberry Pi Pico

Concepts: System Design, REST APIs, OOP, Agile, NLP, Speaker Diarization

Achievements

- Compressed CV model from **45 MB to 12 MB** with real-time inference.
- Built full local ML workflows using RTX 3060 with **zero cloud cost**.
- Benchmarked FAISS+Llama-8B against ChromaDB+Llama-70B for RAG latency and retrieval quality.
- Authored IEEE-style research documentation for FireSmokeNet.
- Led real-time EV Digital Twin visualization platform.